

Cedric Phillips

+1 604 809 8259 | cedric.phillips@mail.mcgill.ca | [LinkedIn](#) | [qaqqaqaaq.github.io](https://github.com/qaqqaqaaq)

Montreal, QC, Canada

EDUCATION

- **McGill University** Fall 2022 - Winter 2026 (expected)
B.Sc. Honours Applied Mathematics Montreal, Canada
Minor concentrations: GIS & Remote Sensing, Hindi Language
 - GPA: 3.60/4.00
 - Winter 2025: Exchange at the École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
 - Courses: Graduate probability theory & mathematical stats., numerical analysis, partial differential equations, stochastic processes, algorithm design.
- **Politecnico di Milano** Starting Fall 2026
M.Sc. Mathematical Engineering, Applied Statistics track Milan, Italy

EXPERIENCE

- **McGill University** Winter 2025
Grader, MATH 458: Honours Differential Geometry Funded by TEAM
 - Verified solutions to assigned problems and returned assignments to all students with detailed feedback in a consistent, timely manner as the sole grader for the course

TECHNICAL SKILLS & PROFICIENCIES

- **Programming/data analysis:** Python (polars, NumPy, matplotlib), R (qrmtools), SQL, Java
- **Mathematical:** $\text{\LaTeX}$, Mathematica, Lean prover
- **Languages:** English (native), French (native), Hindi-Urdu (advanced)

PROJECTS

- **Honours Research Project** Winter 2026
Hierarchical copula dependence models Supervised by Prof. Nešlehová
 - Developing ways to infer the tree structure of a hierarchical model for multivariate Bernoulli variables from Kendall and Spearman rank correlation matrix data
 - Generating and transforming samples from the model using R and investigating applications of the model to credit risk, culminating in a final report
- **Stochastic optimization in asymmetric games** Present-April 2026
Maximal scoring strategies for the Lizard Cult in Root To appear in [McGill's undergraduate math journal](#)
 - Created a Markov chain model for the Root game state and the Lizard Cult's play options; analyzed decision-making under uncertainty variables to find moves that optimize expected value
 - Validated results of project via Monte Carlo simulation, produced visualizations of scoring trajectories with parameter sensitivity using Python
- **Directed Reading Project** Summer 2024
Proof formalization in the Lean language Supervised by Dr. Turcotte
 - Members of our project group studied the underlying type theory to learn the Lean language from scratch
 - Explored areas of personal mathematical interest, learning advanced proof techniques as necessary, culminating in an original constructive Lean-verification of the Cauchy-Schwarz inequality in $\mathbb{R}^n$
- **Nonlinear Dynamics Honours Project** Fall 2023
Canards and other explosive behaviours Supervised by Prof. Humphries
 - Independently conducted research on a chosen graduate-level topic: canard 'explosions', Hopf bifurcations, and the 'blow-up' map in slow-fast dynamical systems
 - Read literature to produce a referenced report summarizing the most important insights & applications

WORK

- **Wild Blue Restaurant & Bar, Caramba Restaurant** Summers and winters, 2021-2026
Expeditor Whistler, BC
 - Logistics under pressure: quality-checked all dishes being sent out of the kitchen, organized a non-stop flow of orders & the resources needed to send them out